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Abstract

GRASP (greedy randomized adaptive search procedures) is a multi-start meta-heuristic designed to find high-quality solutions for combinatorial optimization problems. Each iteration of GRASP typically consists of two phases: a construction phase, where a feasible solution is built, and a local search phase, which begins from the constructed solution and performs iterative improvements until a locally optimal solution is achieved. While the construction phase is

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usually based on a randomized greedy algorithm, other construction methods have also been explored. Repeated use of the construction procedure generates diverse starting solutions for the local search. This chapter provides an overview of GRASP, outlining its fundamental components and various enhancements to the basic approach, such as Reactive GRASP, intensification strategies, and random-key GRASP.

Keywords

GRASP · Continuous GRASP · Metaheuristics

Introduction

In a combinatorial optimization problem, given a finite (or countably infinite) solution set X and a real-valued objective function $f : X \rightarrow \mathbb{R}$, the goal is to find a solution $x^* \in X$ such that $f(x^*) \leq f(x)$ for all $x \in X$. While some of these problems can be solved in polynomial time, many are computationally intractable, as no exact polynomial-time algorithms are currently known to solve them [72]. Moreover, most real-world problems are either inherently computationally intractable or too large for exact algorithms to be practical. In these cases, metaheuristic methods are commonly used to find good, though not necessarily optimal, solutions. These methods typically alternate between phases of intensification and diversification. Intensification refers to focusing the search in a promising area of the solution space to refine and improve the current solution, often by applying more thorough local search techniques. On the other hand, diversification aims to explore new and potentially unexplored regions of the solution space, preventing the search from becoming stuck in local optima. Over time, various heuristic search techniques have been developed that effectively balance these two phases, significantly improving our ability to find high-quality solutions to complex combinatorial optimization problems. These techniques include simulated annealing [119], tabu search [73], genetic algorithms [101], biased random-key genetic algorithms [104], variable neighborhood search [85], adaptive large neighborhood search [151], and GRASP (greedy randomized adaptive search procedures) [52, 53].

GRASP is a multi-start metaheuristic designed to find good-quality solutions for combinatorial optimization problems. Each iteration consists of a construction phase, where a solution (which can be feasible or infeasible) is built, followed by a local search phase. The local search begins at the constructed solution and iteratively improves it until a locally optimal solution is reached. Although the construction phase is typically based on a randomized greedy algorithm, other construction procedures have been proposed. Repeated applications of the construction procedure generate diverse starting solutions for the local search, and the best local optimum found across all iterations is returned as the final solution.

This chapter overviews GRASP by describing its basic components along with some among the most fruitful proposed enhancements, including reactive GRASP,

Table 1 Applications of GRASP: operations research problems

Routing	[17, 20, 24, 32, 36, 95, 137]
Logic	[45, 50, 70, 121, 132, 136]
Covering and partition	[15, 16, 52, 74, 83, 118, 130]
Location	[1, 42–44, 60, 93, 150]
Minimum Steiner tree	[33, 108, 109, 141]
Optimization in graphs	[2, 19, 35, 55, 59, 99, 120, 133, 148]
Assignment	[5, 51, 71, 102, 112, 122, 123, 135, 145]
Timetabling and scheduling	[4, 7, 11, 12, 25, 29, 54, 56, 64, 81, 96, 105, 139, 142–144]

Table 2 Applications of GRASP: some industrial applications

Manufacturing	[6, 22, 23, 30, 94, 115]
Transportation	[17, 21, 51, 63, 147]
Telecommunications	[9, 10, 41, 93, 103, 125, 128, 149]
Graph and map drawing	[68, 69, 98, 107, 116, 133]
Power systems	[27, 28, 49]
Computational biology	[65, 79, 87]
VLSI	[14, 15]

intensification strategies, and hybridization with other metaheuristics. The chapter is organized as follows. Basic components, alternative construction mechanisms, and local search characteristics are described in the next section. Enhancements to the basic procedure, including reactive GRASP and intensification strategies, are discussed in section “**Enhancements.**” Section “**Hybridizations**” describes several state-of-the-art hybridizations of GRASP with other metaheuristics, while section “**Continuous GRASP**” describes how GRASP has been extended to continuous optimization problems. Tables 1 and 2 report a number of GRASP implementations that have appeared in the literature, covering a wide range of applications in several and heterogenous fields. Moreover, the reader can refer to [66, 67, 134].

Basic Components

Given a finite solution set X and a real-valued objective function $f : X \rightarrow R$ to be minimized, a basic GRASP metaheuristic [52, 53] is a multi-start or iterative method, in which each iteration consists of two phases: construction of a solution and local search.

The construction phase builds a solution x , usually feasible, but optionally also infeasible. If x is not feasible, a repair procedure may be invoked to obtain feasibility. Once a solution x is obtained, its neighborhood is investigated by the

```

algorithm GRASP( $f(\cdot)$ ,  $g(\cdot)$ , MaxIterations, Seed)
1   $x_{best} := \emptyset$ ;  $f(x_{best}) := +\infty$ ;
2  for  $k = 1, 2, \dots, \text{MaxIterations} \rightarrow$ 
3     $x := \text{ConstructGreedyRandomizedSolution}(\text{Seed}, g(\cdot))$ ;
4    if ( $x$  not feasible) then
5       $x := \text{repair}(x)$ ;
6    endif
7     $x := \text{LocalSearch}(x, f(\cdot))$ ;
8    if ( $f(x) < f(x_{best})$ ) then
9       $x_{best} := x$ ;
10   endif
11 endfor;
12 return ( $x_{best}$ );
end GRASP

```

Fig. 1 Pseudo-code of a basic GRASP for a minimization problem

local search until a local minimum is found. The best overall solution is kept as the result. The pseudo-code in Fig. 1 illustrates the main blocks of a GRASP procedure for minimization, in which MaxIterations iterations are performed and Seed is used as the initial seed for the pseudorandom number generator.

Construction Phase

Starting from an empty solution, a complete solution is iteratively constructed in the construction phase, one element at a time (see Fig. 2). At each construction iteration, the choice of the next element to be added is determined by ordering all candidate elements (i.e., those that can be added to the solution) in a candidate list C with respect to a greedy function $g : C \rightarrow R$. This function measures the (myopic) benefit of selecting each element. The heuristic is adaptive because the benefits associated with every element are updated at each iteration of the construction phase to reflect the changes brought on by the selection of the previous element. The probabilistic component of a GRASP is characterized by randomly choosing one of the best candidates in the list, but not necessarily the top candidate. The

```

procedure ConstructGreedyRandomizedSolution(Seed,  $g(\cdot)$ )
1    $x := \emptyset$ ;
2   Sort the candidate elements  $i$  according to their incremental costs  $g(i)$ ;
3   while ( $x$  is not a complete solution)  $\rightarrow$ 
4        $\text{RCL} := \text{MakeRCL}()$ ;
5        $v := \text{SelectIndex}(\text{RCL}, \text{Seed})$ ;
6        $x := x \cup \{v\}$ ;
7       Resort remaining candidate elements  $j$  according to  $g(j)$ ;
8   endwhile;
9   return( $x$ );
end ConstructGreedyRandomizedSolution;

```

Fig. 2 Basic GRASP construction phase pseudo-code

list of best candidates is called the *restricted candidate list* (RCL). This choice technique allows for different solutions to be obtained at each GRASP iteration but does not necessarily compromise the power of the adaptive greedy component of the method.

The most used technique to build the RCL applies the min max- α percentage rules. At any GRASP iteration, let $g_{\min}$ and $g_{\max}$ be the smallest and largest incremental costs, respectively, i.e.,

$$g_{\min} = \min_{i \in C} g(i), \quad g_{\max} = \max_{i \in C} g(i). \quad (1)$$

The RCL is made up of elements $i \in C$ with the best (i.e., the smallest) incremental costs $g(i)$. There are two main mechanisms to build this list: *cardinality-based* (CB) and *value-based* (VB) mechanisms. In the CB case, the RCL is made up of the k elements with the best incremental costs, where k is a parameter. In the VB case, the RCL is associated with a parameter $\alpha \in [0, 1]$ and a threshold value $\mu = g_{\min} + \alpha(g_{\max} - g_{\min})$. All candidate elements i whose incremental cost $g(i)$ is no greater than the threshold value are inserted into the RCL, i.e., $g(i) \in [g_{\min}, \mu]$. Note that the case $\alpha = 0$ corresponds to a pure greedy algorithm, while $\alpha = 1$ is equivalent to a random construction. The pseudo-code in Fig. 3 is a refinement of the greedy randomized construction pseudo-code shown in Fig. 2.

Prais and Ribeiro in [126, 127] analyzed GRASP with different RCL construction mechanisms, varying the parameter α in three ways:

```

procedure ConstructGreedyRandomizedSolution(Seed,  $\alpha$ ,  $k$ ,  $g(\cdot)$ )
1   $x := \emptyset$ ;
2  Initialize the candidate set  $C$  by all elements;
3  Evaluate the incremental cost  $g(i)$  for all  $i \in C$ ;
4  while ( $|C| > 0$ )  $\rightarrow$ 
5       $g_{min} := \min_{i \in C} g(i)$ ;  $g_{max} := \max_{i \in C} g(i)$ ;
6      if (CB RCL is used) then
7          Sort candidate elements  $i \in C$  according to  $g(i)$ ;
8           $RCL := C[1 \dots k]$ ;
9      else  $RCL := \{i \in C \mid g(i) \leq g_{min} + \alpha(g_{max} - g_{min})\}$ ;
10     endif;
11      $v := \text{SelectIndex}(RCL, \text{Seed})$ ;
12      $x := x \cup \{v\}$ ;
13     Update the candidate set  $C$ ;
14     Reevaluate the incremental costs  $g(i)$  for all  $i \in C$ ;
15 endwhile;
16 return( $x$ );

end ConstructGreedyRandomizedSolution;

```

Fig. 3 Refined pseudo-code of the GRASP construction phase

- (a) α randomly chosen from a uniform distribution
- (b) α randomly chosen from a decreasing nonuniform distribution
- (c) Fixed α , close to the greedy choice

They applied these strategies to four optimization problems, and they found that strategy (c) had the shortest computation times and less variability, while strategy (a) achieved more frequent hits, showing the effectiveness of varying α .

Additionally, they tested a self-adjusting version of GRASP, where α is modified based on solution quality. This extension is known as Reactive GRASP and will be discussed further in the next section.

Local Search Phase

As is the case for many deterministic methods, the solutions generated by a GRASP construction are not guaranteed to be locally optimal with respect to simple neighborhood definitions. Hence, it is almost always beneficial to apply a local search to attempt to improve each constructed solution. A local search algorithm works in an iterative fashion by successively replacing the current solution by a better solution in the neighborhood of the current solution. It terminates when no better solution is found in the neighborhood. The *neighborhood structure* N for a problem relates a solution s of the problem to a subset of solutions $N(s)$. A solution s is said to be *locally optimal* if there is no better solution in $N(s)$ with respect to the objective function value. The key to success for a local search algorithm consists of the suitable choice of a neighborhood structure, efficient neighborhood search techniques, and the starting solution. Figure 4 illustrates the pseudo-code of a generic local search procedure for a minimization problem.

In a GRASP framework, a local search starts from an initial solution $x_0 \in X$ and iteratively generates a sequence of improving solutions $x_1, \dots, x_M$, where $M = \text{MaxIterations}$. At the k -th iteration, $k = 1, \dots, M$, x_k is locally optimal with respect to the neighborhood $N(x_{k-1})$ since $N(x_{k-1})$ is searched for an improving solution x_k such that $f(x_k) < f(x_{k-1})$. If such a solution is found, it is made the current solution. Otherwise, the search ends with x_{k-1} as a local optimum.

The effectiveness of local search depends on several factors, such as the neighborhood structure, the function to be minimized, and the starting solution. It has been experimentally shown that randomly generated solutions are of poor quality on average. On the other hand, greedy algorithms usually produce solutions of better quality than those of randomly generated solutions. Therefore, using

```

procedure LocalSearch( $x, f(\cdot)$ )
1  Let  $N(x)$  be the neighborhood of  $x$ ;
2   $H := \{y \in N(x) \mid f(y) < f(x)\}$ ;
3  while ( $|H| > 0$ )  $\rightarrow$ 
4       $x := \text{Select}(H)$ ;
5       $H := \{y \in N(x) \mid f(y) < f(x)\}$ ;
6  endwhile
7  return( $x$ );
end LocalSearch

```

Fig. 4 Pseudo-code of a generic local search procedure

greedy solutions as starting points for local search in a multi-start procedure will usually lead to good, though, most often, suboptimal solutions. This is because the amount of variability in greedy solutions is small, and it is less likely that a greedy starting solution will be in the basin of attraction of a global optimum than a random solution. A greedy randomized construction as the one embedded in GRASP adds variability to the greedy algorithm.

Enhancements

To improve the performance of the basic GRASP framework, most efforts have focused on construction mechanisms. Since Mockus et al. [114] pointed out that GRASP with a fixed nonzero RCL parameter α is not asymptotically convergent to a global optimum (During construction, a fixed RCL parameter may rule out a candidate that is present in all optimal solutions.), several remedies have been proposed to get around this problem. They include reactive GRASP, cost perturbations in place of randomized selection, bias functions, memory and learning, and local search on partially constructed solutions.

Reactive GRASP

The results of the study conducted in [126, 128] involving variation of the value of the RCL parameter α motivated the proposal of the extension of the basic GRASP called reactive GRASP. Prais and Ribeiro in [128] have shown that using a single fixed value for the value of RCL parameter α very often hinders finding a high-quality solution, which eventually could be found if another value was used. Moreover, one drawback of the basic GRASP is the lack of *learning* from the history of solutions found in previous iterations. The basic algorithm discards information about any solution encountered that does not improve the incumbent. Instead, it is worth to use information gathered from good solutions leading to *memory-based* procedures. Information about the quality of previously generated solutions can influence the construction phase, by modifying the selection probabilities associated with each element of the RCL.

In this paragraph, we describe reactive GRASP, the first enhancement that incorporates a learning mechanism in the memoryless construction phase of the basic GRASP. In reactive GRASP, the value of the RCL parameter α is selected in each iteration from a discrete set of possible values with a probability that depends on the solution values found along the previous iterations. One way to accomplish this is to use the rule proposed in [128]. Let $\mathcal{A} = \{\alpha_1, \alpha_2, \dots, \alpha_m\}$ be the set of possible values for α . At the first GRASP iteration, all m values have the same probability to be selected, i.e.,

$$p_i = \frac{1}{m}, \quad i = 1, 2, \dots, m. \quad (2)$$

At any subsequent iteration, let $\hat{z}$ be the incumbent solution and let A_i be the average value of all solutions found using $\alpha = \alpha_i$, $i = 1, \dots, m$. The selection probabilities are periodically reevaluated as follows:

$$p_i = \frac{q_i}{\sum_{j=1}^m q_j}, \quad (3)$$

where $q_i = \hat{z}/A_i$, $i = 1, \dots, m$. If values of $\alpha = \alpha_i$ ($i \in \{1, \dots, m\}$) lead to the best solutions on average, then the value of q_i is increased and larger values of q_i correspond to more suitable values for α . The probabilities associated with these more appropriate values will then increase when they are reevaluated.

Due to greater diversification and less reliance on parameter tuning, reactive GRASP has been successfully applied to several problems and has led to improvements over the basic GRASP in terms of robustness and solution quality [8, 26, 146].

Cost Perturbations

Another step toward an improved and alternative solution construction mechanism is to allow *cost perturbations*. The idea is to introduce some “noise” in the original costs in a fashion that resembles the noising method of Charon and Hudry [38, 39]. Cost perturbations are effective in all cases when the construction algorithm is not very sensitive to randomization, as, for example, in the case of the Steiner problem in graphs [62].

Another situation where cost perturbations can be effective is when there is no greedy algorithm available for the problem to be solved, as, for example, in the case of the prize-collecting Steiner tree problem. To solve this problem, the hybrid GRASP procedure proposed by Canuto et al. in [34] used the primal-dual algorithm of Goemans and Williamson [80] to build initial solutions using perturbed costs. More in detail, in [34], at each iteration a new solution for the prize-collecting Steiner tree problem is built using node prizes updated by a perturbation function, according to the structure of the current solution. Two different prize perturbation schemes are used to enforce search diversification, as described in the following.

Perturbation by eliminations: The primal-dual algorithm used in the construction phase is driven to build a new solution without some of the nodes appearing in the solution constructed in the previous iteration. This is done by changing to zero the prizes of some persistent nodes, which appeared in the last solution built and remained at the end of the local search. A parameter μ controls the fraction of the persistent nodes whose prizes are temporarily set to zero.

Perturbation by prize changes: Similarly to the noising method of Charon and Hudry [38, 39], some noise is introduced into the node prizes, resulting in a change of the objective function as well. For each node i , a perturbation factor $\beta(i)$ is randomly generated in the interval $[1 - a, 1 + a]$, where a is

an implementation parameter. The original prize $\pi(i)$ associated with node i is temporarily changed to $\pi(i) = \pi(i) \cdot \beta(i)$.

Experimental results have shown that embedding a strategy of costs perturbation into a GRASP framework improves the best overall results. The hybrid GRASP proposed for the Steiner problem in graphs by Ribeiro et al. in [141] uses this cost perturbation strategy and is effective.

Bias Functions

Another construction mechanism was proposed by Bresina [31]. Once built the RCL, instead of choosing with equal probability one candidate among the RCL elements, Bresina introduced a family of probability distributions to bias the selection toward some particular candidates. A bias function is based on a rank $r(x)$ assigned to each candidate x according to its greedy function value and is evaluated only for the elements in RCL. Several different bias functions were introduced:

- i. Random bias: $\text{bias}(r(x)) = 1$
- ii. Linear bias: $\text{bias}(r(x)) = 1/r(x)$
- iii. Log bias: $\text{bias}(r(x)) = \log^{-1}[r(x) + 1]$
- iv. Exponential bias: $\text{bias}(r(x)) = e^{-r}$
- v. Polynomial bias of order n : $\text{bias}(r(x)) = r^{-n}$

Let $\text{bias}(r(x))$ be one of the bias functions defined above. Once these values have been evaluated for all elements of the RCL, the probability p_x of selecting element x is

$$p_x = \frac{\text{bias}(r(x))}{\sum_{y \in \text{RCL}} \text{bias}(r(y))}. \quad (4)$$

A successful application of Bresina's bias function can be found in [29], where experimental results show that, although valid on all candidates, the evaluation of bias functions may be restricted only to the elements of the RCL.

Biased randomized GRASP (BR-GRASP) is a similar approach proposed in [61]. It applies a nonuniform, skewed probability distribution to the element selection process during the solution construction phase, but respect to Bresina [31], BR-GRASP allows all potential elements to be considered, with biased probabilities assigned to them. This change helps the algorithm converge more quickly to high-quality solutions, while keeping computational effort and the need for additional parameters minimal. Usually, the bias is given by a geometric distribution with a parameter that skews the probabilities.

Reactive GRASP has been the first and very simple attempt to enhance the basic GRASP in order to save and use history from previous iterations. Another very simple attempt is due to Fleurent and Glover [71] who proposed improved

constructive multi-start strategies that besides defining a special bias function also maintain a pool of *elite solutions* to be used in the construction phase. To become an elite solution, a solution must be either better than the best member of the pool, or better than its worst member and sufficiently different from the other solutions in the pool, in order to preserve not only solution quality but also the diversity of solutions. Fleurent and Glover defined: (1) a *strongly determined variable* as one that cannot be changed without eroding the objective or changing significantly other variables; (2) a *consistent variable* as one that receives a particular value in a large portion of the elite solution set, and (3) for each solution component i , a measure $I(i)$ of its strongly determined and consistent features that becomes larger as i appears more often in the pool of elite solutions. The intensity function $I(i)$ is used in the construction phase as follows. Recall that $g(i)$ is the greedy function, i.e., the incremental cost associated with the insertion of element i into the solution under construction. Let $K(i) = F(g(i), I(i))$ be a function of the greedy and the intensification functions. The idea of Fleurent and Glover is to define a special bias function that depends on $K(\cdot)$. In fact, the intensification scheme biases selection from the RCL to those elements i with a high value of $K(i)$ by setting its selection probability to be

$$p_i = \frac{K(i)}{\sum_{y \in \text{RCL}} K(y)}. \quad (5)$$

They suggested $K(i) = \lambda g(i) + I(i)$, with $K(i)$ varying with time by changing the value of λ , e.g., initially λ may be set to a large value that is decreased when diversification is called for. Rules and procedures for changing the value of λ are given by Fleurent and Glover [71] and Binato et al. [29].

Hybridizations

As enhancements to its basic framework, different hybridizations of GRASP with several other metaheuristics have been studied and proposed in the literature. In this section, some of them are surveyed and briefly described.

Laguna and Gonzalez-Velarde in 1991 [96] have first studied hybridization of GRASP with tabu search. Recently, in [131], GRASP is hybridized with tabu search to solve a variant of the multi-compartment vehicle routing problem (MCVRP). The hybrid approach uses GRASP to construct an initial solution and improves it by the tabu search algorithm, which intensifies the search by exploring the solution space more thoroughly. This combination enables the algorithm to balance exploration and exploitation, improving the overall solution quality by refining the initial GRASP solution while avoiding cycles and enhancing diversification. The proposed hybrid GRASP/TS method outperforms other heuristics in terms of solution quality and computational efficiency for the MCVRP.

GRASP has been used also in conjunction with genetic algorithms. Basically, the feasible solution found by using a GRASP construction phase has been used as

initial population by a genetic algorithm, as, for example, in [18] and in [3], where a greedy genetic algorithm is proposed for the quadratic assignment problem.

Another interesting hybridization of GRASP involves VNS (variable neighborhood search) and variable neighborhood descent (VND) proposed by Hansen and Mladenović [84, 113]. Almost all randomization effort in the basic GRASP algorithm involves the construction phase, while local search stops at the first local optimum. On the other hand, strategies such as VNS and VND rely almost entirely on the randomization of the local search to escape from local optima. With respect to this issue, probabilistic strategies such as GRASP and VNS may be considered as complementary and potentially capable of leading to effective hybrid methods.

A first attempt in the direction of integrating VNS into GRASP was done by Martins et al. [110]. The construction phase of their hybrid heuristic for the Steiner problem in graphs follows the greedy randomized strategy of GRASP, while the local search phase makes use of two different neighborhood structures as a VND strategy. Their heuristic was later improved by Ribeiro, Uchoa, and Werneck [140], one of the key components of the new algorithm being another strategy for the exploration of different neighborhoods. Ribeiro and Souza [138] also combined GRASP with VND in a hybrid heuristic for the degree-constrained minimum spanning tree problem. Festa et al. [69] studied different variants and combinations of GRASP and VNS for the MAX-CUT problem, finding and improving some of the solutions that at the time were the best known solutions for some open instances from the literature. In [57], the authors studied several hybridization of GRASP, including VNS, for the far from most string problem. Recently, Di Puglia Pugliese et al. [46] proposed a hybridization of GRASP with VND to solve a vehicle routing problem with occasional drivers. The VND has been used as local search phase, after a biased randomized greedy construction procedure.

At last, we devote the remainder of this section to the combination of the basic GRASP with path relinking. Path relinking was originally proposed by Glover [75] as an intensification strategy exploring trajectories connecting elite solutions obtained by tabu search or scatter search [76–78]. It can be traced back to the pioneering work of Kernighan and Lin [92]. Starting from one or more elite solutions, paths in the solution space leading toward other guiding elite solutions are generated and explored in the search for better solutions. This is accomplished by selecting moves that introduce attributes contained in the guiding solutions. At each iteration, all moves that incorporate attributes of the guiding solution are analyzed, and the move that best improves (or least deteriorates) the initial solution is chosen.

The first proposal of a hybrid GRASP with path relinking was in 1999 due to Laguna and Martí [97]. It was followed by several extensions, improvements, and successful applications [37, 47, 48, 58, 111, 124, 129]. Path relinking is applied to a pair of solutions $\mathbf{x}$ and $\mathbf{y}$, where one can be the solution obtained from the current GRASP iteration, and the other is a solution from an elite set of solutions. $\mathbf{x}$ is called the *initial solution*, and $\mathbf{y}$ the *guiding solution*. The set $\mathcal{E}$ of elite solutions has usually a fixed size that does not exceed MaxElite . Given the pair $\mathbf{x}, \mathbf{y}$, the procedure starts by computing the symmetric difference $\Delta(\mathbf{x}, \mathbf{y})$ between the two solutions, i.e., the set of moves needed to reach $\mathbf{y}$ (target solution) from $\mathbf{x}$ (initial

solution). A path of solutions is generated linking $\mathbf{x}$ and $\mathbf{y}$. The best solution x^* in this path is returned by the algorithm. Since there is no guarantee that x^* is locally optimal, often local search is applied, starting from x^* , and the resulting locally optimal solution is returned.

Figure 5 illustrates the pseudo-code of the path-relinking procedure applied to the pair of solutions $\mathbf{x}$ (starting solution) and $\mathbf{y}$ (target solution). In line 1, an initial solution $\mathbf{x}$ is selected at random among the elite set of elements, and usually, it differs sufficiently from the guiding solution $\mathbf{y}$. The loop in lines 6 through 14 computes a path of solutions $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_{d(\mathbf{x}, \mathbf{y})-2}$, local search is applied in line 15, and the solution x^* with the best objective function value is returned in line 16.

```

algorithm Path-relinking( $f(\cdot), \mathbf{x}, \mathcal{E}$ )

1  Choose, at random, a pool solution  $\mathbf{y} \in \mathcal{E}$  to relink with  $\mathbf{x}$ ;
2  Compute symmetric difference  $\Delta(\mathbf{x}, \mathbf{y})$ ;
3   $f^* := \min\{f(\mathbf{x}), f(\mathbf{y})\}$ ;
4   $x^* := \arg \min\{f(\mathbf{x}), f(\mathbf{y})\}$ ;
5   $x := \mathbf{x}$ ;
6  while ( $\Delta(x, \mathbf{y}) \neq \emptyset$ )  $\rightarrow$ 
7       $m^* := \arg \min\{f(x \oplus m) \mid m \in \Delta(x, \mathbf{y})\}$ ;
8       $\Delta(x \oplus m^*, \mathbf{y}) := \Delta(x, \mathbf{y}) \setminus \{m^*\}$ ;
9       $x := x \oplus m^*$ ;
10     if ( $f(x) < f^*$ ) then
11          $f^* := f(x)$ ;
12          $x^* := x$ ;
13     endif;
14 endwhile;
15  $x^* := \text{LocalSearch}(x^*, f(\cdot))$ ;
16 return ( $x^*$ );

end Path-relinking

```

Fig. 5 Pseudo-code of a generic path relinking for a minimization problem

This is achieved by advancing one solution at a time in a greedy manner. At each iteration, the procedure examines all moves $m \in \Delta(x, \mathbf{y})$ from the current solution x and selects the one which results in the least cost solution (line 7), i.e., the one which minimizes $f(x \oplus m)$, where $x \oplus m$ is the solution resulting from applying move m to solution x . The best move m^* is made, producing solution $x \oplus m^*$ (line 9). The set of available moves is updated (line 8). If necessary, the best solution x^* is updated (lines 10–13). The procedure terminates when $\mathbf{y}$ is reached, i.e., when $\Delta(x, \mathbf{y}) = \emptyset$, returning the best solution found.

The procedure described above is the greedy PR, and it is just one way to apply path relinking. In the context of GRASP, path relinking (PR) has been applied in various forms, which enhance the search for better solutions. The main types of path relinking are greedy, random, and mixed. Randomized PR introduces a random element in selecting the moves, which helps explore diverse solutions and avoid local optima. Mixed PR combines both strategies, alternating between greedy and random moves during the path construction. This method aims to balance the exploration of the solution space and the exploitation of promising solutions. The choice between these strategies often depends on the specific problem and the desired balance between solution quality and computational effort.

Moreover, the concept of multiple parents introduces a novel strategy where several guiding solutions, rather than just one, are used to generate the path from the initiating solution. This approach creates a broader search trajectory, allowing the relinking process to combine attributes from multiple solutions, potentially leading to better solutions by exploring more diverse areas of the solution space. The path from the initiating solution to the guiding solutions is constructed by incorporating attributes from the guiding set based on their frequency of occurrence, which helps preserve beneficial solution characteristics. Additionally, interior and exterior PRs refer to different ways of managing the solutions during the relinking process. Interior PR focuses on incorporating attributes that are present in the guiding solution but not in the initiating solution, progressively modifying the initiating solution by adding the missing attributes. Exterior PR, in contrast, seeks to introduce elements that are not present in either the initiating or guiding solution, thereby creating more diversified intermediate solutions. This form of PR is particularly useful when diversification is required to avoid premature convergence to local optima. Both approaches aim to enrich the search either by focusing on attributes that are common in high-quality solutions or by exploring more distant solution spaces. Readers interested in further details can explore Laguna et al. [100].

Continuous GRASP

Continuous GRASP (C-GRASP) has been proposed in [86] and is an extension of the traditional GRASP algorithm, adapted for solving continuous global optimization problems. Unlike its discrete counterpart, which operates on combinatorial spaces, C-GRASP targets problems in continuous domains, specifically those constrained by box constraints. The algorithm follows a multi-start approach,

alternating between construction and local improvement phases, similar to the original GRASP method. During each iteration, C-GRASP constructs a solution using a greedy randomized approach, where each component of the solution is determined by evaluating the best possible outcome in its local search space. The local improvement phase refines the solution further by exploring the neighboring region for better solutions without relying on derivative information, making C-GRASP particularly useful for “black-box” problems where derivative-based optimization methods cannot be applied. C-GRASP’s effectiveness lies in its ability to adapt the search resolution based on the progress of the solution, intensifying the search when improvements are slow. This makes it a robust and flexible tool for tackling complex global optimization problems in various fields.

In the construction phase, an initial solution is generated using a greedy randomized approach tailored for continuous optimization. The process begins by selecting a random point within the feasible region defined by the problem’s constraints, such as the lower and upper bounds for each variable. Initially, all the coordinates of the solution are considered “unfixed,” meaning that each dimension of the solution can vary. For each unfixed coordinate, a line search is conducted along that coordinate’s direction while holding the other coordinates fixed. This line search aims to find the value that minimizes the objective function for that coordinate. After performing the line search for all unfixed coordinates, an RCL is created, which includes coordinates associated with the best objective function values. The RCL is determined by selecting those coordinates whose objective function values fall within a specific range, controlled by the parameter α . This parameter balances the level of greediness in the selection process, with higher values of α focusing more on the best options. Once the RCL is formed, a coordinate is randomly selected from it, ensuring some level of diversity in the generated solution. The selected coordinate is then fixed at the value determined from the line search. This process repeats for each coordinate until all dimensions of the solution are fixed, resulting in a complete solution.

The local search begins with the solution obtained from the construction phase. For a given solution, the algorithm explores possible moves by considering different directions from the current point in the solution space. These directions are formed by vectors with components set to values from the set $\{1, 0, -1\}$, indicating movement in the positive, zero, or negative direction along each dimension of the solution. To maintain tractability, a limit is set on the number of directions to explore. The number of directions is determined by a parameter, `MaxDirToTry`, which can either be fixed or based on the dimensionality of the problem. For each direction, the algorithm tests a new point by adjusting the current solution along that direction. If the new point improves the objective function (i.e., it yields a better solution), the search continues from this new point, and the process repeats with the updated solution. The local search continues iterating, modifying the solution along different directions, until no further improvements can be found in the selected directions. If a better solution is found, it becomes the new current best solution. If no improvements are possible after testing all allowed directions, the local search terminates, and the resulting solution is returned. This phase serves to fine-tune the

solution, progressively exploring the local neighborhood and refining the solution quality. The randomness in the direction selection helps avoid getting stuck in local minima by diversifying the search, while the use of multiple direction vectors increases the likelihood of discovering a better solution.

Hirsch et al. [90] proposed several techniques to speeding up C-GRASP, while de Araújo et al. [13] presented several direct search procedures that are used in a C-GRASP heuristic. Moreover, C-GRASP has been successfully applied to various problems, including solving nonlinear equations [89], matching projected 3D points and lines [91], economic dispatch of thermal units [117], and others [82, 88, 106].

An interesting variant of C-GRASP is the random-key GRASP (RK-GRASP) presented by Chaves et al. [40]. It integrates the random-key optimizer (RKO) paradigm to solve combinatorial optimization problems. In RK-GRASP, solutions are represented as vectors of random keys, which are real numbers in the interval $[0, 1)$. The random-key vector encodes the solution, and a decoder is used to map this vector to a feasible solution in the problem space. The RK-GRASP procedure follows the typical two-phase structure of GRASP: the construction phase and the local search phase. In the construction phase, a semi-greedy approach is used to generate a solution. This phase starts by selecting a random-key vector, and then each key is perturbed using a line search technique, ensuring that the solutions evolve based on both greedy choices and some randomness. The algorithm builds the solution by iteratively selecting keys and updating the solution. The local search phase then refines the constructed solution by exploring its neighborhood. Three distinct local search strategies have been used: grid search, Nelder-Mead search, and random variable neighborhood descent (RVND). These strategies explore different neighborhoods of the solution space to improve the solution's quality. RK-GRASP is particularly attractive because it decouples the problem-specific components (the decoder) from the problem-independent solver, allowing for the application of the same algorithm to various combinatorial optimization problems by simply changing the decoder. This makes RK-GRASP a flexible and efficient approach for solving a wide range of optimization problems.

Conclusions

The goal of this chapter was to provide an overview of GRASP describing its basic components and enhancements to the basic procedure, including reactive GRASP, intensification strategies, and its hybridizations with different metaheuristics.

Cross-References

- [Genetic Algorithms](#)
- [Multi-start Methods](#)
- [Variable Neighborhood Descent](#)
- [Variable Neighborhood Search](#)

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